BIF401 **Created by Mehwish**

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Q1 ;Optimal energy function in structural prediction?

Ans: Energy based methods involve evaluating the free energy structures. To compute the RNA sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm. RNA Secondary Structure Prediction. Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing energies (+ve values). And also compute the sum of +ve and –ve energies. All possible 2' structures are generated. The best 2' structure is selected

Q2; RAW file Formet (5)

is a format in which an instrument outputs data in binary form. Several software exist for converting RAW file formats into open software formats. Each open format has its own unique advantages. mzXML and MGF formats are most frequently used

Q;3 Types of secondary structures of RNA:

1. Single stranded: 3'end may fold on to the 5' end

2. Helices: Double stranded RNA helix of stacked base pairs

3. Hairpin loop: The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4**. Bulge Loops**: Bulges, are formed when a double-stranded region cannot form base pairs perfectly

5. Interior loop: Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

Q4Advantage and disadvantage of Ab initio:

Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- - These methods are slow due to the huge possibilities

.Q5 Describe scores in silico fragmentation?

Ans: Silico fragmentation scoring:

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score\
- With “all possible” fragments in in silico spectrum, and “reported” fragments in experimental spectrum, we can match and rank.
- Scoring scheme should also consider the errors in peak matching

- **Q6';Function of Tandem Ms**:

  Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

**Q7Write steps of protein sequence identification? **

Ans: Mass spectrometers are used to measure the molecular weight of proteins and peptides. Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. . ionization by mass spectrometery
4. . fragmentation MS2
5. . mass spectra
6. . EST Determination
7. . filter protein database
8. . in silico fragmentation of candidate protein
9. .matching of experimental and insilco peak list
10. . post translational modification
11. protein score.

Q8Silico fragment scoring?

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Q9HOW Propensity table help us

For a primary sequence, and a tentative 2’ structure, propensity table can help us compute the overall propensity. Product of propensity values is computed for overall propensity for each 2’ structure. An important point to note here is that 2’ structures are formed due to hydrogen bonding between amino acids.

So, we need to consider the neighboring amino acids as well.

You only need to compute propensities for a small number 2’ structures. The highest net propensity will be the most probably secondary structure that will be formed.

Q10 Factors That Participate In Folding Protein\

The external factors involved in protein denaturation or disruption of the native state include temperature, external fields (electric, magnetic), molecular crowding, and even the limitation of space, which can have a big influence on the folding of proteins. (Internet

Q11. Different Secondary Structures Of RNA?

The secondary structures of biological DNA's and RNA's tend to be different: biological DNA mostly exists as fully base paired double helices, while biological RNA is single stranded and often forms complex and intricate base-pairing interactions due to its increased ability to form hydrogen bonds stemming from the

Q12Role of amino acid

These structural properties are equally important in giving rise to protein structures Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein • Also, chemically inactive amino acids reduce chances of destabilizing reactions in core

Q13Briefly describe chou fasman algorithm? \

Ans: Chou – Fasman method: It is a technique for the prediction of secondary structures in proteins i.e. Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived For the appearance of each amino acid in each secondary structure type. To predict the probability that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein

Q14 Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required.\
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops

Q15Use of mass spectrometer

Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides. Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured.

Q16Hydrophobicity of amino acid

mino acids have characteristics like polarity, hydrophobicity, and charge states. These characteristics are governed by the elemental composition of an amino acid's side chain (R group).

HYDROPHILIC AMINO ACIDS Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non-

polar. Hydrophobic amino acids are mostly found at the inside of folded proteins. Hydrophilic group contain the chain of C and H group in their R grouP

Q17Role of Amino Acids

We know that amino acids can be polar, charged and hydrophobic. Role of polar and charged amino acids in folding. Role of hydrophobic amino acids in folding

Q18Overall Goal of Folding

Anfinsen's thermodynamic hypothesis: Proteins fold for a unique, stable and minimum free kinetic energy structure. What other factors may come into play for satisfying Anfinsen hypothesis

Q19Need For Chou Fasman Algorithms?

The Chou-Fasman algorithm is simple in principle.The conformational parameters for each amino acid were calculated by considering the relative frequency of a given amino acid within a protein, its occurrence in a given type of secondary structure, and the fraction of residues occurring in that type of structure.

Q20 **Role of Global Protein**?

Global protein function prediction from protein-protein interaction networks. ... The availability of entire genome sequences and of high-throughput capabilities to determine gene coexpression patterns has shifted the research focus from the study of single proteins or small complexes to that of the entire proteome. (Internet)

Q21Chou – Fasman method?

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Q22. What Is Ab Initio Modelling? Describe The Limitations Of This Model?\\

Ab initio methods have Anfinsen'sthermodynamic hypothesis at the center • These methods attempt to identify the structure with minimum free energy. • Applicable to any sequence • Not very accurate biologically • Accuracy & applicability are limited by our understanding of the protein folding problem. Limitation • Computationally expensive • Suitable for proteins with less than 100 residues (PPT)

.Q23 **What is meant by “PK” value of an amino acid?**

Ans: PK is the value for an amino acid is that PH at which exactly half of amino acids are charged and half are not charged

Limitation: Models can be conceptual, graphical or mathematical as they are used in science. It allows one to break down a concept into simpler terms with a visual component. A limitation of models in science is that they are usually simplified versions of the real situation or concept. (Internet)

Q24 Why folding is important?

Ans: Proteins fold spontaneously. Proteins fold

to achieve thermodynamic stability. Proteins fold to organize themselves for performing functions in cells